

To determine the melting point of a solid substance having low melting point

Video Reference :- <https://www.youtube.com/watch?v=ggXHDZFsQBc>

Materials Required :-

- Thermometer
- 100 ml Beaker
- Tripod Stand
- Stirrer
- Capillary Tube
- Water / Paraffin
- Wire Gauze

Procedure :-

1. Powdering the compound & filling of capillary tube
 - The thin walled capillary tube is sealed from one end by heating it on a gas burner for a few seconds.
 - The given compound is crushed & it is pushed in the capillary tube in powdered form, The tube is repeatedly tapped & filled by the compound.
2. Fix the tube with thermometer
 - The capillary tube is attached to the thermometer.
 - The whole apparatus is positioned in parallel to each other.
 - The beaker in which the tube has to be placed is filled with paraffin or water.
3. Heating the liquid bath
 - The liquid bath is heated gently with continuous stirring so that the constant temperature is maintained throughout the liquid using a glass loop stirrer moving up & down.
 - As the temperature reaches near the melting point of solid the flame is reduced & temperature is allowed to rise gradually.
4. Note the melting point

- Note the temperature when the substance starts melting. The temperature is noted again when the solid melts completely.
- The average or mean of two values gives the correct melting point of the substance.

SI No :	The Temperature When Solid Starts Melting (T_1)	The Temperature When Solidification Starts (T_2)	Average = $T_1 + T_2 / 2$
1	20°C	40°C	30°C
2	40°C	60°C	50°C